AI/NLP Risk Engine
+ Strategic Portfolio Stress Test

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Problem & Approach

Risk and ratings teams can't manually read every headline fast enough to know what it means for a portfolio.

• Built: news + social text → sentiment/event/impact signals → a real portfolio stress test.

What we built it with

• Verify every external claim live — never report a number a script didn't actually produce.

• Two sources, one engine: live Mastodon posts and a historical tweet-replay fallback.

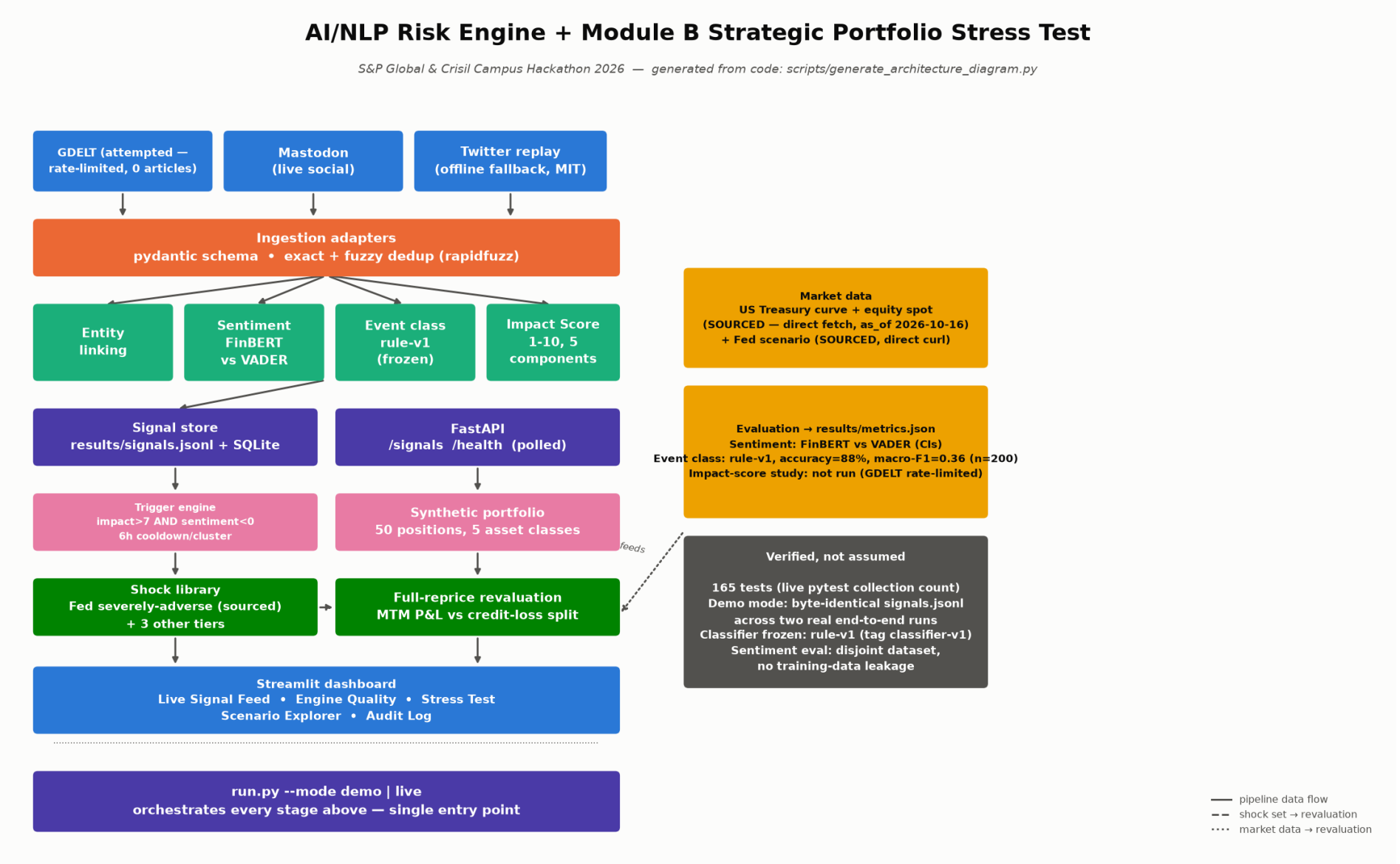
Assumptions that turned out false

• X/Twitter has no free read API in 2026 → switched to Mastodon.

• FinBERT is fine-tuned on Financial PhraseBank → evaluated on a disjoint dataset to avoid leakage.

• Entity linking: a naive name match collided with everyday phrases ("Target" in "price target") → fixed with cashtag matching + context rules.

System Design

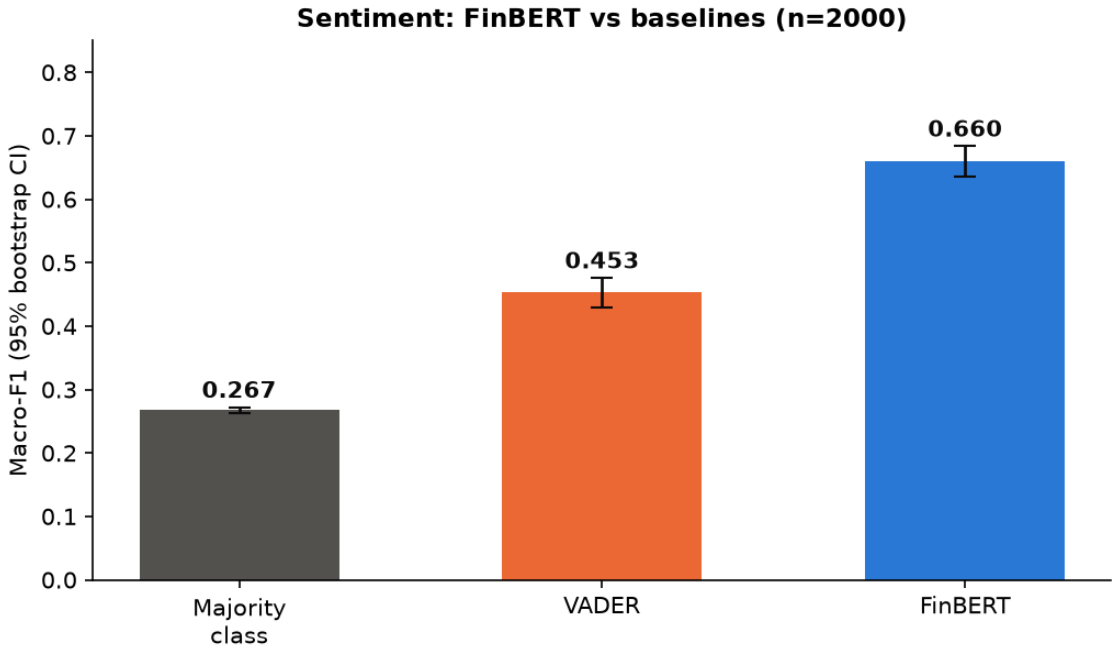


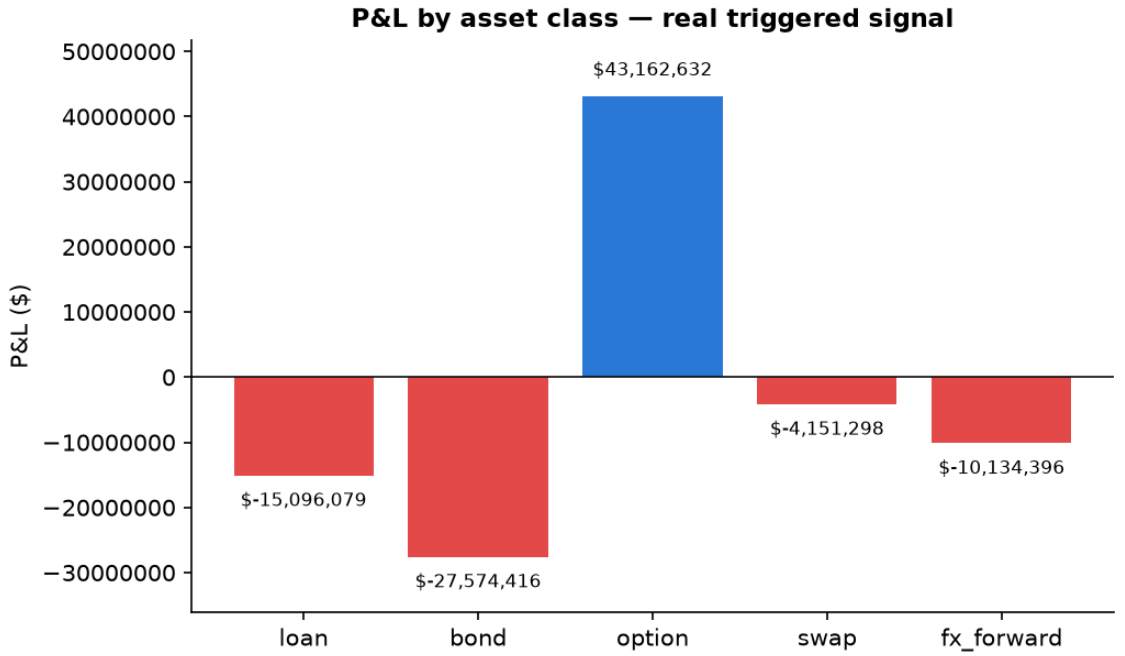
Implementation Highlights

- FinBERT vs VADER vs majority-class — scored with bootstrap 95% CIs, not point estimates alone.
- Event classifier frozen before scoring against hand-labelled gold data — no eval-set tuning.
- Polling-based pub/sub (FastAPI + a flat file) — documented honestly as polling, not a real message bus.
- Full cashflow reprice for bonds, swap DV01 via bump-and-reprice, closed-form Black-Scholes for options — each compared against the required naive approximation, not just implemented once.
- Mark-to-market P&L and credit loss kept as two separate numbers — a position is never both fair-valued and provisioned for the same deterioration.
- Demo mode verified byte-identical across two real end-to-end runs, by a dedicated automated test.

Key Results

- FinBERT 0.660 macro-F1 vs VADER 0.453 vs majority-class 0.267 (n=2000) — bearish recall 76.5%.
- 76 real signals evaluated, 3 fired. Naive reprice error reaches 0.42% of position value — genuine convexity effect.
- Event classifier: 88.5% vs 84.0% always-'Other' baseline; Macroeconomic precision 100%, recall 34%.





Headline result: total stressed loss -$13,793,557 (-2.09%)

= MTM P&L +$1,302,523 + credit loss -$15,096,079

Domain Impact

• Translates headlines into a dollar P&L a risk desk can act on — signal → scenario → shock → P&L, fully traceable.

Mirrors real risk-desk practice

• MTM P&L and credit loss stay separate — a trading-book instrument and a banking-book loan are never collapsed into one number.

• Shocks anchored to the Fed's own severely-adverse scenario — sourced, not arbitrary round numbers.

• Taxonomy spans credit events, rating-adjacent language, portfolio stress — the surface a ratings/risk-analytics business operates on. (Credit-event detection itself isn't validated yet.)

$13.79M stressed loss (-2.09%)

Fed severely-adverse scenario on the synthetic portfolio, triggered by a real headline

Limitations & Next Steps

Evaluation status

- Impact-score event study: not run — GDELT rate-limited (0 articles over 2+ hours), abandoned for this submission.
- Event classifier: 88.5% vs 84.0% always-'Other' baseline; Macroeconomic precision 100%, recall 34%.
- Credit Event: too few examples to evaluate (n=1).
- Geopolitical: too few examples to evaluate (n=2).

Known limitations & next steps

- Market data is SOURCED, not provisional — fetched directly.
- English-only; GDELT is headline-only; the social replay source has synthetic timestamps.
- Credit loss is 1-year EL, not IFRS 9 lifetime — a lower bound, stated as such.
- Next: grow the Credit Event / Geopolitical samples; investigate the Macroeconomic recall gap.